ESHA HOODA

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Seeking Summer 2027 SWE & ML/AI Internships (+ possess PM skills)

EDUCATION

BSc. Data Science, Minors in Statistics & Computer Science, San Jose State University Dec 2027
Coursework: Machine Learning, Artificial Intelligence, Statistics & Probability, Data Visualization, Data Structures & Algorithms, Database Management, Advanced Python, OOP, Linear Algebra, Discrete Math, Computer Vision, R, Social Network Analysis
Certificate (San Jose State University): Fundamental AI Algorithms and Applications in Computer Science (expected May 2027)

SKILLS & CERTIFICATIONS

Languages: Python, SQL, JavaScript, Java, C++ | **ML & Data:** PyTorch, TensorFlow/Keras, HuggingFace Transformers, Scikit-learn, XGBoost, OpenCV, PySpark, Pandas, NumPy, ETL | **AI Engineering:** Anthropic Claude API (tool-calling, agent design), RAG, Claude Code, GitHub Copilot | **Software Engineering:** Flask, FastAPI, MySQL, Git, UNIX | **Cloud & DevOps:** GCP, BigQuery, Docker, Airflow, dbt | **Data Viz:** Tableau, Power BI, Matplotlib, Seaborn | **Product & Project:** roadmapping, sprint planning, A/B testing
Certifications: Python, Machine Learning, Artificial Intelligence (IIT Bombay & IIT Madras), Code in Place (Stanford), IRB Research Training, CRLA Level 1 & 2

TECHNICAL EXPERIENCE

Deep Learning Intern - National University of Singapore & Hewlett Packard Enterprise Dec 2022 - Jan 2023
- Designed and implemented **ANN, RNN, GAN, SVM, NLP, and time series** models and deployed them through **Azure** pipelines
- Trained a posture detection model by tuning neural network architecture and iterating on experiments, reaching **93% accuracy**

AI Upskilling Research Program, Stanford SALT Lab Sep 2026
- Building a **RAG** document-search system and tool-using AI agent, hardening both against attack, with a **CI** eval regression gate and a model server deployed into a private cloud network

Technical Learning Assistant - Peer Connections, SJSU Jan 2025 - Jan 2026
- Mentored **250+ students** in math and CS and co-designed equity-focused in-class activities with professors; CRLA Level 1 & 2 certified

RESEARCH EXPERIENCE

Machine Learning Research Assistant, SJSU (Dr. Cassandra Paul) May 2025 - Present
- Conducting independent research with **Python (pandas, NumPy, scikit-learn), OLS, logistic regression, hierarchical linear modeling (HLM), and propensity score matching** on **~70K student records** quantifying the LA model's impact on DFW rates, graduation rates, and retention (gender equity focus); journal paper manuscript underway
- Quantifying the underlying biases (instructor effect, course effect, student effect, etc.) that inform all LA impact literature

Machine Learning Research Assistant, Diffusion Models (Dr. Tahir Bachar Issa) 2026
- Applying **diffusion models (SR3, image super-resolution via iterative refinement)** in place of GANs for video-rate fluorescence microscopy super-resolution, building on an IEEE BIBE 2020 GAN paper

PROJECT EXPERIENCE

LegalBee: Jurisdiction-Aware Legal Information Copilot (SJSU Responsible Computing Club) Sep 2026
- Leading a **team of 7**, setting roadmap and sprint priorities, delegating workstreams, and coordinating weekly check-ins
- Developing a jurisdiction-aware **RAG** copilot with citations, tracking retrieval accuracy and hallucination rate

Daybeam: AI Day-Planning Agent Oct 2026
- Defined success rate, cost, and speed metrics to compare **3 planning strategies** on **hundreds** of test tasks
- Built an agent with **tool calls** to an **A*** route planner, places lookup, and constraint checker, plus a **Streamlit** app

Trust & Safety Risk Pipeline Jul 2026
- Scoped one risk score across **3 signal types**, validated by **A/B test** against a baseline
- Combined **XGBoost, HuggingFace/PyTorch, OpenCV** on a **PySpark, Airflow, dbt, BigQuery** pipeline served via **FastAPI/Docker**

Fake News Detection Using LSTM & BERT May 2026
- Compared **4 architectures**, flagged a Reuters source confound, and recommended cross-source validation
- Trained **SVM, Random Forest, BiLSTM** (Keras), and **BERT** (HuggingFace) on **20,000 articles**, reaching **99.9% F1**

Stock Volatility Prediction Pipeline May 2026
- Found prior-day range the top volatility predictor (**0.837**) and COVID volatility **50-80%** above baseline
- Built a **PySpark and SQL** pipeline on **GCP** over **600,000+ daily trades across 500+ stocks** with **RandomForest** and partitioned **Parquet** output

LEADERSHIP EXPERIENCE

Outreach Lead - Peer Connections, SJSU May 2025 - Jan 2026
- Scaled a major initiative from 0 to **3,000 participants in 2 months** by redesigning internal processes and building **SQL**-based tracking systems while coordinating a team of **120+**